F.2. SVCCA at Alternative Thresholds

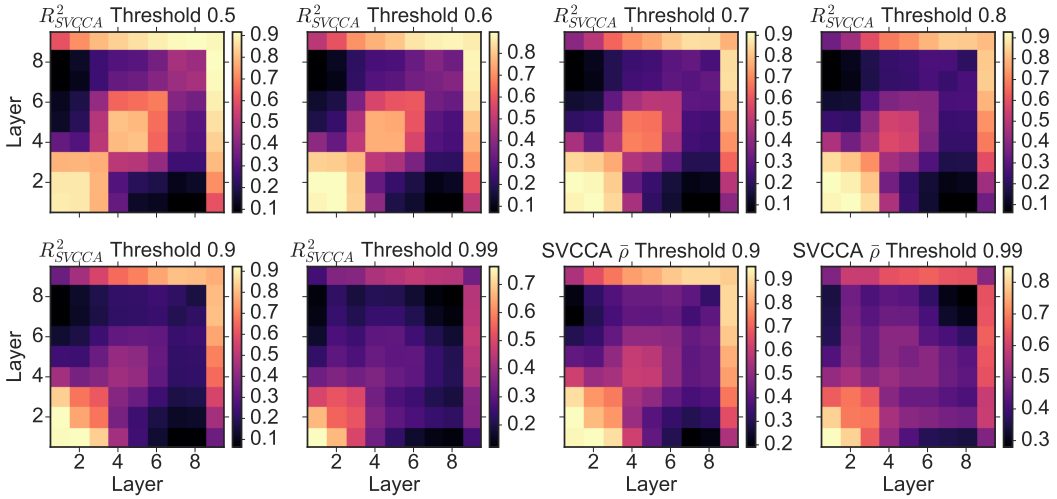

Figure F.3. Same as Figure 2 row 2, but for more SVCCA thresholds than the 0.99 threshold suggested by Raghu et al. (2017). No threshold reveals the structure of the network.

F.3. CKA at Initialization

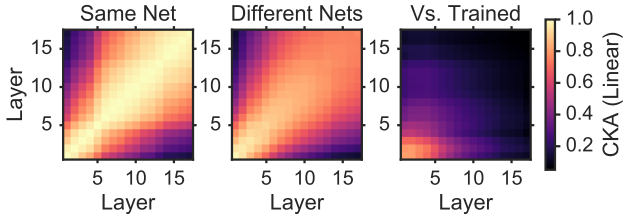

Figure F.4. Similarity of the Plain-18 network at initialization. **Left**: Similarity between layers of the same network. **Middle**: Similarity between untrained networks with different initializations. **Right**: Similarity between untrained and trained networks.

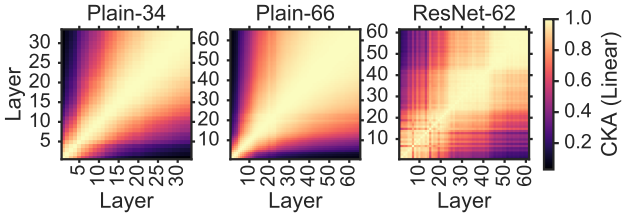

Figure F.5. Similarity between layers at initialization for deeper architectures.

F.4. Additional CKA Results

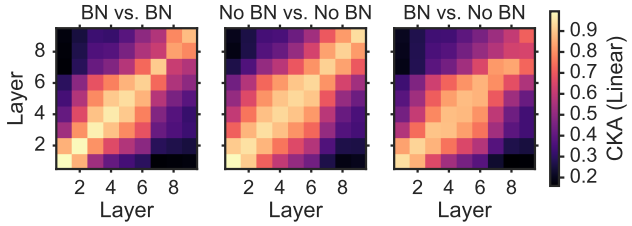

Figure F.6. Networks with and without batch normalization trained from different random initializations learn similar representations according to CKA. The largest difference between networks is at the last convolutional layer. Optimal hyperparameters were separately selected for the batch normalized network (93.9% average accuracy) and the network without batch normalization (91.5% average accuracy).

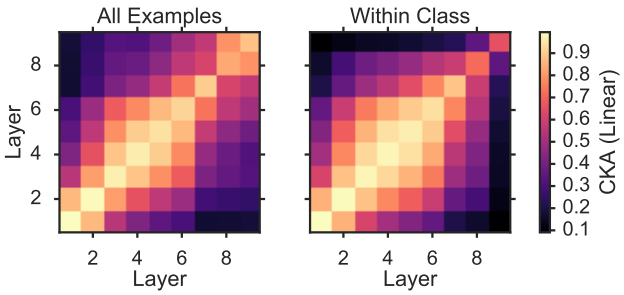

Figure F.7. Within-class CKA is similar to CKA based on all examples. To measure within-class CKA, we computed CKA separately for examples belonging to each CIFAR-10 class based on representations from Plain-10 networks, and averaged the resulting CKA values across classes.

F.5. Similarity Between Different Architectures with Other Indexes

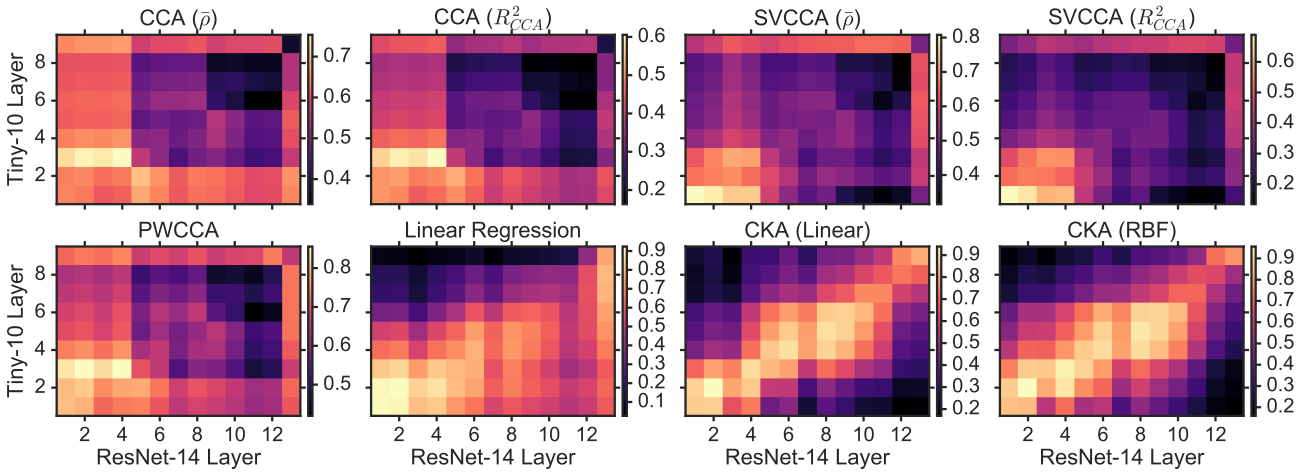

Figure F.8. Similarity between layers of different architectures (Tiny-10 and ResNet-14) for all methods investigated. Only CKA reveals meaningful correspondence. SVCCA results resemble Figure 7 of Raghu et al. (2017). In order to achieve better performance for CCA-based techniques, which are sensitive to the number of examples used to compute similarity, all plots show similarity on the CIFAR-10 training set.